

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years of experience** designing and owning cloud-native, high-throughput distributed systems.
- Expertise in **Python and Go microservices**, event-driven architectures, real-time and batch data processing, **Agentic AI and RAG systems**.
- Proven track record shipping **production systems**, leading backend migrations, and improving reliability, performance, and scalability.

SKILLS

- Languages:** Python, Go (Golang), C, C++, SQL.
- Backend & APIs:** FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, WebSockets, SSE, Celery.
- AI & Agents:** LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, text-to-SQL, semantic caching, pgvector, ChromaDB.
- Streaming & Data:** Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, ClickHouse, BigQuery, DuckDB.
- Databases & Storage:** PostgreSQL, MySQL, Redis, MongoDB, Cassandra, Elasticsearch, Amazon S3.
- Cloud & Infrastructure:** GCP, AWS, Docker, Kubernetes, Git, CI/CD, Observability (ELK, Grafana, New Relic, Sentry).
- Core Engineering:** Data Structures & Algorithms, System Design (HLD/LLD), Concurrency, DBMS, Operating Systems, Unit Testing.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore June 2026 – Present

Agentic AssortSmart (AI Powered Retail Merchandise Planning Platform)

- Building the agentic rebuild of AssortSmart, a merchandise planning SaaS where retailers decide what to buy, how much, and for which stores, with AI agents drafting store clusters and plans for planner review, override, and approval.
- Developing the **agentic microservice (FastAPI, LangGraph)** with a conversational agent that grounds plain language requests into store and season scopes, batch evaluates **100 silhouette scored candidate clusterings**, and presents the **top 3 with evidence** for human approval, cutting turnaround from days to **under 1 hour**.
- Designing the core backend as **Go (Gin) microservices** covering plan lifecycle, tenant configuration, and bulk save REST APIs at **~10–20 peak RPS**, with goroutine worker pools, bounded channels, context timeouts, and JWT auth middleware.
- Led the design of the per tenant **ClickHouse** planning store on an **append only, never erase write model** at **~5–10K row writes/sec**. Facts land as atomic partition swaps; decisions are inserts via latest state views for version diff, undo/redo, lock free writes, and a full audit trail.
- Tech Used:** Python, FastAPI, LangGraph, MCP, Go, Gin, ClickHouse, PostgreSQL, Redis, Kafka, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Cut Uber's quarterly financial risk-scoping cycle time by **70% (~2 weeks to ~3–4 days)** by replacing a Google Sheets workflow with a FastAPI/MySQL platform serving **36 REST endpoints across 8 screens (p95 < 300ms)**, feeding PwC audit work papers.
- Designed an **8-table normalized MySQL schema** with a quarterly ETL that aggregates **19M+ general-ledger rows into 300K rows**, reconciled against public 10-Q filings.
- Migrated raw SQL access to **SQLAlchemy 2.0 ORM**, fixing a latent column-aliasing bug and raising changed-module test coverage to **100%**.
- Tech Used:** Python, FastAPI, SQLAlchemy, Pydantic, MySQL, Redis, Bazel, pytest.

Menu Ingestion & Automation Platform (Uber Eats)

- Built an event-driven pipeline with **Kafka (~200–500 peak events/sec)**, **Flink**, and **Spark** for **30,000+ vendor menus/month**, cutting onboarding from 24h to 2h and saving **\$600K+/yr**.
- Built an AI extraction pipeline with **RAG and Gemini** converting PDF and image menus into schema-validated catalog data with **98% fidelity**.
- Increased successful ingestions by **95% (~60–65% to 95%+)** via IP rotation and proxy pools; health dashboards on **Apache Pinot** (sub-second).
- Tech Used:** Python, Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, Selenium, Gemini, GCP, Docker.

Software Development Engineer 2 MASTERS INDIA, Noida December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Led the migration from a PHP (Laravel) monolith to asynchronous **Python (FastAPI) microservices**, mentoring junior engineers through the rollout and cutting p95 API latency from **1.2s to 300ms** for 2,500+ enterprise clients.
- Scaled bulk e-invoice processing to **100,000+ transactions per import and 1M+ daily transactions (~12 TPS avg, 100+ TPS peak)** with async workers and idempotent retries.
- Set up **ELK and New Relic alerting**, cutting incident triage by **70% (~30 min to <10 min)**, and raised test coverage from **35% to 82%** with pytest gating CI/CD.
- Tech Used:** Python, FastAPI, PostgreSQL, Redis, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer GEEKSFORGEEKS, Noida August 2021 – November 2022

- Redesigned and migrated the backend from PHP to **Django (Python)**, improving reliability for **100,000+ daily queries (~1–2 RPS avg, ~10× spikes)**.
- Designed data models and REST APIs for voting and pinning, lifting **premium subscriptions 15–20%** relative.
- Tech Used:** Python, Django, MySQL, Redis, REST, Cron, AWS.

EDUCATION

B.Tech (Information Technology) Institute of Engineering & Technology, Lucknow July 2017 – June 2021

ACHIEVEMENTS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier Round 2021.
- Finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.